

Juan Irving Vasquez Gomez



Date of birth: November, 1984

Position: Professor at Instituto Politécnico Nacional (IPN), Mexico. Centro de Innovación y Desarrollo Tecnológico en Cómputo (CIDETEC) <https://www.cidetec.ipn.mx/>

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Research Interests:

Robotics, Computer Vision, Deep learning, 3D Modelling, Motion Planning.

• PERSONAL STATEMENT:

I am a scientific researcher passionate about understanding the motion planning algorithms that involve active sensing. My training as an engineer and scientist has allowed me to propose, analyze and implement state-of-the-art solutions for several theoretical and practical problems. My current research interests include deep learning-based computer vision, robot motion planning, autonomous 3D reconstruction, and autonomous surface inspection.

• EDUCATION:

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|------|---|
| 2014 | PhD. in Computer Sciences. <i>Institute for Astrophysics Optics and Electronics (INAOE)</i> 🔗. Thesis: View Planning for 3D Object Reconstruction with Mobile Robots 📄. Advisors: Enrique Sucar and Rafael Murrieta. |
| 2009 | M. Sc. in Computer Sciences. <i>Institute for Astrophysics Optics and Electronics (INAOE)</i> 🔗. Thesis: View Planning for Three-dimensional Object Reconstruction 📄. Advisors: Enrique Sucar and Efrain Lopez-Damian. |
| 2006 | B.S.E. in Computer Systems Engineering <i>Tehuacan Institute of Technology (ITT)</i> , Graduated by "Score of excellence". |

• PROFESSIONAL EXPERIENCE:

- | | |
|----------------|---|
| 2023 - Current | Full Professor. Centro de Innovación y Desarrollo Tecnológico en Cómputo (CIDETEC). Instituto Politécnico Nacional (IPN). |
| 2021 - 2023 | Assistant Professor. Centro de Innovación y Desarrollo Tecnológico en Cómputo (CIDETEC). Instituto Politécnico Nacional (IPN). |

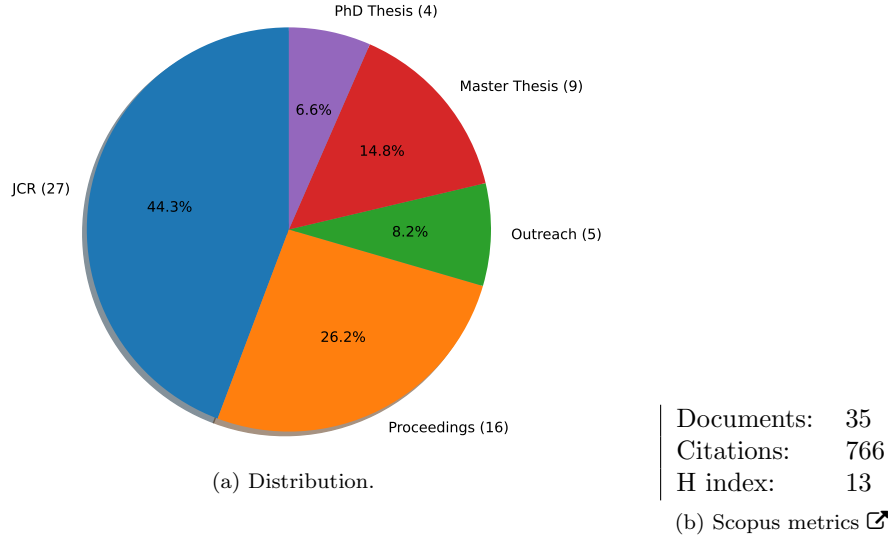


Figure 1: Publication's distribution and impact.

02/2016 - 2021	CONACYT Research Fellow. assigned to Instituto Politécnico Nacional (IPN), Centro de Innovación y Desarrollo Tecnológico en Cómputo (CIDETEC). Project 1507.
08/2009 - 08/2014	Researcher and programmer for the INAOE's Markovito Team at the Robocup@home Competition.

• PUBLICATIONS:

– JCR Indexed Journals:

- * Flores-Aquino, Gabriel Omar and Gutiérrez-Frías, Octavio and Vasquez-Gomez, Juan Irving, Path Planning using a One-shot-sampling Skeleton Map, *IEEE Latin America Transactions*, (2026), , I.F. 1.3
- * David Ernesto Troncoso Romero and Miriam Guadalupe Cruz Jiménez and Juan Irving Vasquez-Gomez and Massimiliano Laddomada and Uwe Meyer-Baese, Adding Reconfiguration Capabilities to the Cascaded Integrator-Comb (CIC) Decimation Architecture, *Accepted to IEEE Signal Processing Magazine, Tips & Tricks*, (2026), , I.F. 10.3
- * Gerardo Antonio Alvarez Hernandez, Juan Irving Vasquez, Abril Valeria Uriarte Arcia, Luis Alberto Tovar-Ortiz, Synthetic Dataset Generation for Tomato Ripening Stage Detection in Different Scenes, *Accepted to IEEE Latin America Transactions*, (2026), I.F. 1.3
- * Edgar Mauricio Muñoz-Silva, Juan Irving Vasquez-Gomez, Carlos Alejandro Merlo-Zapata, Mayra Antonio-Cruz, Binary damage classification of built heritage with a 3D neural network, *npj Heritage Science*, (2025), , I.F. 2.6
- * Garrido-Castañeda, Sergio Isahí and Vasquez, Juan Irving and Antonio-Cruz, Mayra, Coverage Path Planning Using Actor-Critic Deep Reinforcement Learning, *Sensors*, (2025), , I.F. 3.4
- * Olguín-Rojas, Juan Carlos and Vasquez, Juan Irving and López-Canteñs, Gilberto de Jesús and Herrera-Lozada, Juan Carlos and Mota-Delfin, Canek, A Lightweight YOLO-Based Architecture for Apple Detection on Embedded Systems, *Agriculture*, (2025), , I.F. 3.3

- * Zamora Suárez, Ángel Eduardo and Alvarez Hernandez, Gerardo Antonio and Vasquez, Juan Irving and Taud, Hind and Uriarte-Arcia, Abril Valeria and Zamora, Erik, Automatic Correction of Labeling Errors Applied to Tomato Detection, *Agriculture*, (2025), , I.F. 3.6
- * Aldo Ramírez-Arellano and Edgar Mauricio Muñoz-Silva and Mayra Antonio-Cruz and Juan Irving Vasquez-Gomez, Deng entropy and LSTM neural network to classify built cultural heritage with severe damage, *Journal of Cultural Heritage*, (2025), , I.F. 3.5
- * Erick Rodriguez Hernandez, Juan Carlos Olguin Rojas, Gerardo Antonio Alvarez Hernandez, Juan Irving Vasquez-Gomez, Abril Valeria Uriarte Arcia, Hind Taud, Comparing Human and Machine Clustering for Tomato Ripening Stage Classification, *Ciencia e Agrotecnología*, (2024), , I.F. 1.2
- * Mwata-Velu, Tat'y and Zamora, Erik and Vasquez-Gomez, Juan Irving and Ruiz-Pinales, Jose and Sossa, Humberto, Multiclass Classification of Visual Electroencephalogram Based on Channel Selection, Minimum Norm Estimation Algorithm, and Deep Network Architectures, *Sensors*, (2024), , I.F. 3.4
- * Flores-Aquino, Gabriel O. and Vasquez-Gomez, J. Irving and Gutierrez-Frias, O. Octavio, Custom Distribution for Sampling-Based Motion Planning, *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, (2022), , I.F. 2.220
- * Vazquez-Carmona, E. Viridiana and Vasquez-Gomez, J. Irving and Herrera Lozada, Juan Carlos and Antonio-Cruz, Mayra, Coverage Path Planning for Spraying Drones, *Computers and Industrial Engineering*, (2022), , I.F. 5.431
- * Vasquez, Juan Irving and Uriarte-Arcia, Abril Valeria and Taud, Hind and García-Florian, Andrés and Ventura-Molina, Elías, Coastal Sargassum Level Estimation from Smartphone Pictures, *Applied Sciences*, (2022), , I.F. 2.838
- * Rodríguez-Hernandez, Erick; Vasquez, Juan Irving; Duchanoy, Carlos; Taud, Hind, Unsupervised Driving Situation Detection in Latent Space for Autonomous Cars, *Applied Sciences*, (2022), , I.F. 2.679
- * Juan Carlos Olguin-Rojas, Juan Irving Vasquez, Gilberto Castens, CLASIFICACIÓN DE MANZANAS CON REDES NEURONALES CONVOLUCIONALES, *Revista Fitotecnía Mexicana*, (2022), , I.F. 0.418
- * Vasquez, Juan Irving and Merchán-Cruz, Emmanuel Alejandro, A Divide and Conquer Strategy for Sweeping Coverage Path Planning, *Energies*, (2022), , I.F. 3.252
- * Vasquez-Gomez, J Irving and Troncoso, David and Becerra, Israel and Sucar, Enrique and Murrieta-Cid, Rafael, Next-best-view regression using a 3D Convolutional Neural Network, *Machine Vision and Applications*, (2021), , I.F. 1.605
- * Vasquez-Gomez, Juan Irving and Marciano-Melchor, Magdalena and Valentin, Luis and Herrera-Lozada, Juan Carlos, Coverage Path Planning for 2D Convex Regions, *Journal of Intelligent and Robotic Systems*, (2020), , I.F. 2.020
- * Vazquez-Carmona, Viridiana and Vasquez-Gomez, Juan Irving and Herrera-Lozada, Juan Carlos, Environmental Monitoring using Embedded Systems on UAVs, *IEEE Latin America Transactions*, (2020), , I.F. 0.804
- * Mendoza, Miguel and Vasquez-Gomez, J Irving and Taud, Hind and Sucar, Luis Enrique and Reta, Carolina, Supervised Learning of the Next-Best-View for 3D Object Reconstruction, *Pattern Recognition Letters*, (2020), , I.F. 2.810
- * Yervilla-Herrera, Heikel and Vasquez-Gomez, J Irving and Murrieta-Cid, Rafael and Becerra, Israel and Sucar, L Enrique, Optimal motion planning and stopping test for 3-D object reconstruction, *Intelligent Service Robotics*, (2019), , I.F. 1.346
- * Lopez-Jimenez, Efren and Vasquez-Gomez, Juan Irving and Sanchez-Acevedo, Miguel Angel and Herrera-Lozada, Juan Carlos and Uriarte-Arcia, Abril Valeria, Columnar cactus recognition in aerial images using a deep learning approach, *Ecological Informatics*, (2019), , I.F. 2.310

- * Olguin-Carbajal, M and Herrera-Lozada, J.C. and Sandoval-Gutierrez, J. and Vasquez-Gomez J.I., and Serrano-Talamantes J.F. and Chavez-Estrada F.A. and Rivera-Zarate, I. and Hernandez-Bolanos, M., A Micro-DE Algorithm for Continuous Complex Functions, *IEEE Access*, (2019), , I.F. 4.098
- * Vasquez-Gomez, J Irving and Sucar, L Enrique and Murrieta-Cid, Rafael and Herrera-Lozada, Juan-Carlos, Tree-based search of the next best view/state for three-dimensional object reconstruction, *International Journal of Advanced Robotic Systems*, (2018), , I.F. 1.223
- * Olguin Carbajal, Mauricio and Herrera-Lozada, Juan Carlos and Rivera-Zarate, Israel and Serrano-Talamantes, J Felix and Cadena-Martínez, Rodrigo and Vasquez-Gomez, J Irving, Minimum Addition Chains Generation Using Evolutionary Strategies, *Computacion y Sistemas*, (2018), , I.F. 0.100
- * Vasquez-Gomez, J Irving and Sucar, L Enrique and Murrieta-Cid, Rafael, View/state planning for three-dimensional object reconstruction under uncertainty, *Autonomous Robots*, (2017), , I.F. 2.244
- * Vasquez-Gomez, J Irving and Sucar, L Enrique and Murrieta-Cid, Rafael and Lopez-Damian, Efrain, Volumetric next-best-view planning for 3d object reconstruction with positioning error, *International Journal of Advanced Robotic Systems*, (2014), , I.F. 0.526

– **Conferences:**

- * Yair Eduardo Brito Téllez, Sergio Isahí Garrido-Castañeda, Yenny Villuendas-Rey, Juan Irving Vasquez, Comparison of U-Net and FPN Architectures for Ground-Level Semantic Segmentation of sargassum in the Mexican Caribbean, *COMIA 2026, 15th Mexican Conference on Artificial Intelligence*, 2026
- * Bryan Jhoan Cazáres Leyva, Ulises Gachuz Davila, José Juan González Fonseca, Juan Irving Vasquez, Vanessa A. Camacho-Vázquez, and Sergio Isahí Garrido-Castañeda, Interpretable Human Activity Recognition for Subtle Robbery Detection in Surveillance Videos, *MCPR 2026, 15th Mexican Conference on Pattern Recognition*, 2026
- * Vasquez-Gomez, Juan Irving and Troncoso Romero, David E. and Antonio-Cruz, Mayra and Zamora, Erik, Spiral Trajectories for Building Inspection with Quadrotors, *2022 International Conference on Unmanned Aircraft Systems (ICUAS)*, 2022
- * Gabriel O. Flores-Aquino; Jheison Duvier Díaz Ortega; Ricardo Yahir Almazan Arvizu; Raúl López Muñoz; O. Octavio Gutierrez-Frias; J. Irving Vasquez-Gomez, 2D Grid Map Generation for Deep-Learning-based Navigation Approaches, *2021 International Conference on Mechatronics, Electronics and Automotive Engineering (ICMEAE)*, 2021
- * Edgar Mauricio Munoz-Silva, Mayra Antonio-Cruz, Juan Irving Vasquez-Gomez and Carlos Alejandro Merlo-Zapata, A Survey on Point Cloud Generation for 3D Scene Reconstruction, *2021 International Conference on Mechatronics, Electronics and Automotive Engineering (ICMEAE)*, 2021
- * Juan Carlos Olguín-Rojas, J. Irving Vasquez-Gomez, Gilberto de Jesus Lopez-Cantens and Juan Carlos Herrera-Lozada, Un método no invasivo para la clasificación de manzanas, *Research in Computing Science, Congreso Mexicano de Inteligencia Artificial COMIA 2021*, 2021
- * Vasquez-Gomez, J. Irving, VPL: A view planning library for 3D object reconstruction, *International Conference on Mechatronics, Electronics and Automotive Engineering 2020 (ICMEAE)*, 2020
- * Rodríguez-Hernandez, Erick and Vasquez-Gomez, Juan Irving and Herrera-Lozada, Juan Carlos, Flying through Gates using a Behavioral Cloning Approach, *2019 International Conference on Unmanned Aircraft Systems (ICUAS)*, 2019

- * Vasquez-Gomez, Juan Irving and Herrera-Lozada, Juan-Carlos and Olguin-Carbajal, Mauricio, Coverage path planning for surveying disjoint areas, *2018 International Conference on Unmanned Aircraft Systems (ICUAS)*, 2018
- * Vasquez-Gomez, Juan Irving and Herrera-Lozada, Juan Carlos and Olguin-Carbajal, Mauricio, Spatial resolution optimization for terrain coverage with UAVs, *2017 International Conference on Mechatronics, Electronics and Automotive Engineering (ICMEAE)*, 2017
- * Vasquez-Gomez, Juan Irving and Melchor, Magdalena Marciano and Lozada, Juan Carlos Herrera, Optimal coverage path planning based on the rotating calipers algorithm, *2017 International Conference on Mechatronics, Electronics and Automotive Engineering (ICMEAE)*, 2017
- * Vasquez-Gomez, J Irving and Gomez-Castañeda, Cecilia and De Cote, Enrique Muñoz and Herrera-Lozada, Juan Carlos, Multirotor uav coverage planning under wind conditions, *2016 International Conference on Mechatronics, Electronics and Automotive Engineering (ICMEAE)*, 2016
- * Vasquez-Gomez, J Irving and Sucar, L Enrique and Murrieta-Cid, Rafael, View planning for 3d object reconstruction with a mobile manipulator robot, *2014 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2014
- * Vasquez-Gomez, J Irving and Sucar, L Enrique and Murrieta-Cid, Rafael, Hierarchical ray tracing for fast volumetric next-best-view planning, *2013 International Conference on Computer and Robot Vision (CRV)*, 2013
- * Vasquez, Juan Irving and Sucar, L Enrique, Next-best-view planning for 3d object reconstruction under positioning error, *Mexican International Conference on Artificial Intelligence (MICA)*, 2011
- * Vasquez-Gomez, Juan Irving and Lopez-Damian, Efraín and Sucar, Luis Enrique, View planning for 3D object reconstruction, *2009 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2009

– **Preprints:**

- * Gerardo Antonio Alvarez Hernandez, Juan Irving Vasquez, Abril Valeria Uriarte Arcia, Luis Alberto Tovar-Ortiz, Synthetic Dataset Generation for Tomato Ripening Stage Detection in Different Scenes, Submitted to IEEE Latin America Transactions, 2025, [\[A\]](#)

• **STUDENTS:**

– Graduated PhD students:

- * Esther-Viridiana Vazquez Carmona, *Diseño de un algoritmo de cobertura de espacios tridimensionales para vehículos aéreos no tripulados*, [\[A\]](#), 2023, Instituto Politécnico Nacional.
- * Flores Aquino, Gabriel Omar, *Planificación de trayectorias para robots móviles con estrategias de aprendizaje máquina*, [\[A\]](#), 2023, Instituto Politécnico Nacional.
- * Olguin Rojas, Juan Carlos, *Sistema mecatrónico para la clasificación y detección de daños en manzana (*Malus domestica*)*, [\[A\]](#), 2023, Instituto Politécnico Nacional.
- * Erick Rodríguez Hernández, *Navegación autónoma para vehículos en pistas cerradas con obstáculos móviles*, [\[A\]](#), 2022, Instituto Politécnico Nacional.

– Graduated master students:

- * Brito Tellez, Yair Eduardo, *Prototipo de segmentación semántica en la nube para imágenes ambientales*, [\[A\]](#), 2025, Instituto Politécnico Nacional.
- * Casas García, Nadia Montserrat, *Segmentación semántica de imágenes urbanas mediante redes neuronales profundas*, [\[A\]](#), 2025, Instituto Politécnico Nacional.

- * Gante Diaz, Saulo Abraham, *Mapeo implícito autónomo de ambientes interiores* , , 2023, Instituto Politécnico Nacional.
- * Álvarez Hernandez, Gerardo Antonio, *Sistema de visión computacional aplicado a la detección de tomates* , , 2023, Instituto Politécnico Nacional.
- * Muñoz Silva, Edgar Mauricio, *Implementación de un algoritmo de aprendizaje máquina para clasificar una nube de puntos de un modelo arquitectónico* , , 2023, Instituto Politécnico Nacional.
- * Rodriguez Hernandez, Erick, *Clonación de comportamiento para cruce de pasajes estrechos con VANT* , , 2019, Instituto Politécnico Nacional.
- * Vazquez-Carmona, Viridiana, *Sistema electronico para el monitoreo de gases de efecto invernadero utilizando internet de las cosas y vehiculos aereos no tripulados* , , 2019, Instituto Politécnico Nacional.
- * Jimenez, Efren Lopez, *Sistema embebido para la supervision inteligente de terrenos con vehiculos aereos no tripulados* , , 2018, Instituto Politecnico Nacional.
- * Mendoza Guadarrama, Miguel, *NBV-Net: Una red neuronal para calcular la siguiente mejor vista* , , 2018, Instituto Politécnico Nacional.

• CERTIFICATIONS

- 2018, Flying Car Nanodegree, UDACITY, San Francisco California, USA.
- 2018, Computer Vision Nanodegree, UDACITY, San Francisco California, USA.
- 2024, Fundamentals of Accelerated Computing with CUDA Python, NVIDIA, CDMX.